

MUHAMMAD ZAIN

AI ENGINEER

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SUMMARY

AI Engineer with 1+ years of experience specializing in Enterprise Agentic Systems, LLMs, and Computer Vision. Currently at HOFT, architecting a production-grade Multi-Agent software for SAP Invoice Automation using FastAPI and Gemini API. Previously at Qonkar Technologies, experienced in LLMs, RAG, NLP, and Computer Vision. Built Qonkar Chatbot, DocumentGPT, AI Patient Data Analysis System, Fashion Image Search, and Urdu Roman-Urdu Caption Generator using LangChain, Gemini API, OpenAI API, YOLOv11, mT5 (QLoRA), and CLIP, developing POC and prototype applications in Streamlit to validate and refine AI-driven solutions.

PROFESSIONAL EXPERIENCE

AI Engineer, HOFT

Dec 2025 - Present

- Architected an Enterprise-Grade Agentic Invoice Automation system using FastAPI and Gemini API, automating SAP GRN-to-Invoice validation and reducing manual data entry workload by ~80%.
- Developed a scalable Multi-Agent Orchestration Engine to classify and validate complex billing scenarios (One-to-Many, Foreign/Local), utilizing Pydantic for strict type enforcement and data integrity.
- Built a robust SAP Integration Layer for real-time 3-way matching, duplicate detection, and auto-posting, ensuring 100% data consistency across Hadeed, Ittefaq, and Foreign service lines.

AI Engineer, Qonkar Technologies

Dec 2024 - Present

- Engineered and deployed Qonkar Chatbot, a production-grade AI assistant built with LangChain, Gemini API, improving client response time by 30%.
- Deployed DocumentGPT with RAG, automating document understanding at 92% accuracy and reducing manual review efforts by 50%.
- Created Fashion Image Search, leveraging YOLOv11, SAM, and CLIP with FAISS for real-time outfit similarity retrieval.
- Fine-tuned mT5 (QLoRA) for Urdu Roman-Urdu caption generation, integrating with a multilingual Whisper + Seq2Seq audio-text pipeline.

AI Engineer Intern, Qonkar Technologies

Sep 2024 - Dec 2024

- Built a Streamlit POC for the Qonkar Chatbot, exploring LLM frameworks, RAG workflows, and vector database integration to validate prototype concepts.
- Contributed to the development of internal AI tools and evaluated model performance across different LLM APIs and architectures.

PROJECTS

Urdu to Roman-Urdu Caption Generation using mT5 (QLoRA Fine-Tuning)

Sep 2025 - Oct 2025

- Fine-tuned Google mT5-base model using QLoRA on 150K Urdu Roman-Urdu sentence pairs for caption translation.
- Reduced training loss from 0.29 → 0.23 and validation loss from 0.20 → 0.16 across two epochs.
- Utilized 4-bit quantization, LoRA adapters, and efficient checkpointing for memory-optimized training on Kaggle T4 GPU.

- Prepared the fine-tuned model for integration in an AI audio-to-text pipeline combining Whisper and Seq2Seq models.

Fashion Image Search with AI

Aug 2025 - Sep 2025

- Fine-tuned YOLOv11 on the DeepFashion2 dataset to detect and localize clothing items.
- Applied SAM for background removal and generated embeddings using CLIP.
- Implemented FAISS vector search for efficient retrieval of visually similar outfits.
- Built an interactive Streamlit web app enabling real-time fashion image search for e-commerce.

AI-Driven Patient Data Analysis for Enhanced GP Efficiency

Aug 2024 - Jul 2025

- Built a web-based system using RAG and Generative AI to process multi-modal patient data (text, voice, images) for faster, accurate diagnostics.
- Created a synthetic dataset of 300+ simulated consultations and fine-tuned an LLM with doctor-specific expertise.
- Developed an interactive Streamlit UI integrated with LangChain, Hugging Face, Gemini API, and vector databases for real-time clinical support.

DocumentGPT

Mar 2025 - May 2025

- Developed DocumentGPT, an AI-powered tool using Streamlit for interactive document Q&A over PDF and DOCX files.
- Implemented semantic search using HuggingFace embeddings stored in a Qdrant vector database.
- Integrated LangChain with Google's Gemini 2.0 Flash to enable real-time chat with document content.

Web-Based AI Chatbot for Qonkar Technologies

Dec 2024 - Apr 2025

- Built and deployed a fully responsive AI chatbot for Qonkar Technologies using HTML5, CSS3, and vanilla JavaScript.
- Integrated the chatbot to automate customer support, enhance user engagement.
- Applied NLP and LLM techniques to power conversational intelligence without relying on third-party tools.

SKILLS

- **Programming Languages:** Python, JavaScript
- **Frameworks & Libraries:** Scikit-learn, TensorFlow, Keras, PyTorch, NumPy, Pandas, Hugging Face Transformers & Diffusers, Django.
- **Computer Vision Models:** YOLOv11, SAM, CLIP
- **AI & ML Techniques:** Computer Vision, Generative AI, RAG, Vector Databases (FAISS, Qdrant)
- **LLM Tools:** LangChain, LangGraph, OpenAI API, Gemini API
- **Databases:** SQL(PostgreSQL), NoSQL
- **APIs:** REST API
- **Large Language Models:** GPT-4, Gemini
- **Solutions:** Custom Chatbots, Document Q&A Systems, Agentic AI Workflows
- **Soft Skills:** Problem Solving, Communication, Critical Thinking, Team Collaboration

EDUCATION

Bachelor of Engineering in Computer Systems Engineering

Dec 2021 - Jul 2025

- Quaid-e-Awam University of Engineering, Sciences and Technology.

CERTIFICATIONS & AWARDS

- Google AI Essentials - Google | 2025
- Google Prompting Essentials - Google | 2025
- AI Agents Fundamentals – Hugging Face | 2025
- Generative AI for Data Scientists Analytics Specialization – Manthan Patel | 2024